Lab - Investigate BIOS or UEFI Settings

Introduction

In this lab, you will boot the computer, explore the firmware setup utility program, and change the boot order sequence.

Recommended Equipment

- Computer with or without operating system
- Motherboard manual

Instructions

Part 1: Enter BIOS or UEFI.

Step 1: Power on the computer.

a. Plug the power supply cable into an AC wall outlet.

b. If there is a power switch on the power supply, set the switch to “1” or “on”.

c. Turn on the computer with the power button on the front panel.

Note: If the computer beeps more than once, or if the power does not come on, notify your instructor.

Step 2: Enter the firmware setup program.

During POST, press the firmware setup key or key combination. The firmware setup utility program screen will appear.



Questions:

What is the key or combination of keys used to enter the firmware setup utility program?

Delete or F2 can be used.

Who manufactures the BIOS or UEFI system for your computer?

The manufacturer of my BIOS is ASUS.

What is the BIOS or UEFI version?

My BIOS version is 1401.

Part 2: Explore the Settings.

Step 1: List the main menu options.

Question:

List the main menu options and describe what is monitored in each menu?

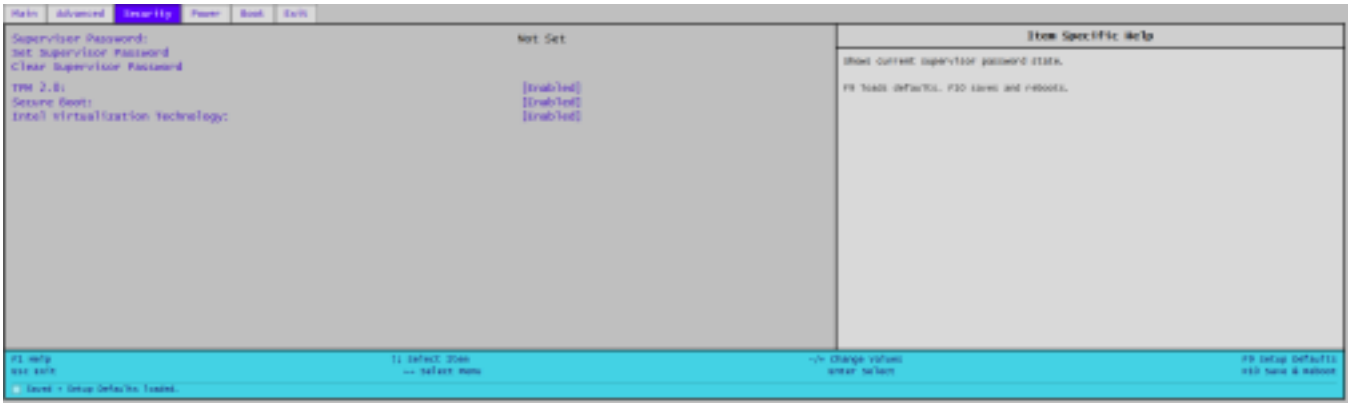


BIOS Version – This is where the BIOS version is located, however it is only a simulation. System Model – This provides the model of the computer system just like the BIOS version. (simulated) Processor Type – This is where the CPU model, brand and type is located.
System Memory – This option has the RAM model and slots corresponding to it.

System Time – It monitors the precise system time of the computer based on the set location. System Date – It contains the date today which is always updated as time goes by.

Step 2: Find the security settings.

Navigate through each screen to find the security settings.



Question:

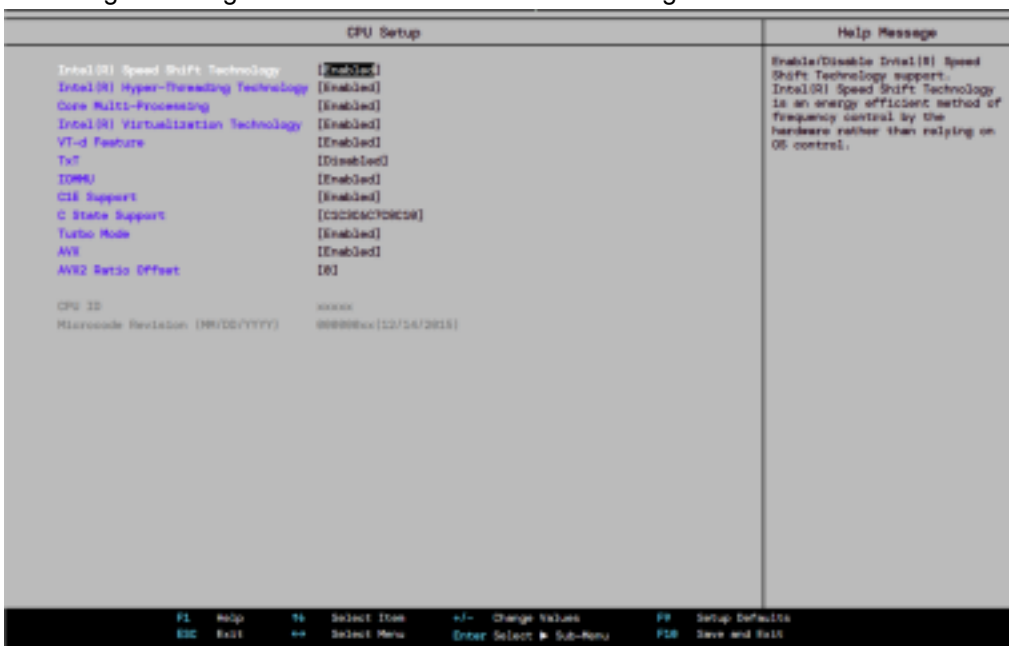
What security settings and features are available?

Supervisor Password – It contains the password to access the BIOS.

Set Supervisor Password – This option is selected after clearing a password of a supervisor. Clear Supervisor Password – This option is for clearing a already present supervisor password. TPM 2.0 – This is the Trusted Platform Module, it ensures integrity of the important keys and files upon boot. Secure Boot – This ensures that only trusted malware softwares open during startup. Intel Virtualization Technology – This provides the PC ability to run multiple operating systems.

Step 3: Find the CPU settings.

Navigate through each screen to find the CPU settings.



Questions:

What is the CPU speed?

The CPU speed is 2.4Ghz.

What other information is listed for the CPU?

Other information include IVT, and features to enhance performance of computer including overclocking.

Step 4: Find the RAM settings

Navigate through each screen to find the RAM settings.



Questions:

What is the RAM speed?

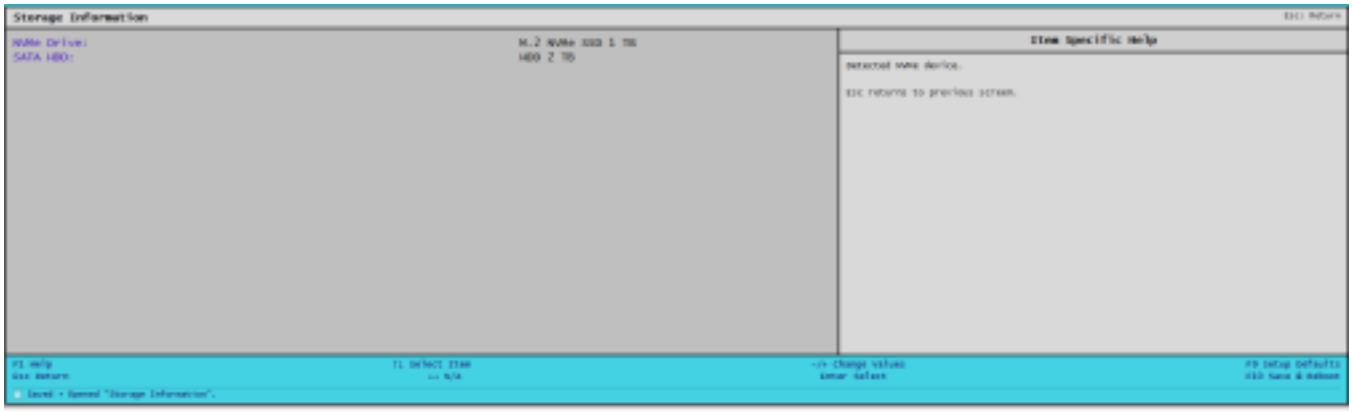
My RAM speed is 2400Mhz.

What other information is listed for the RAM?

Other information include RAM capacity and slot.

Step 5: Find the hard drive settings.

Navigate through each screen to find the hard drive settings.



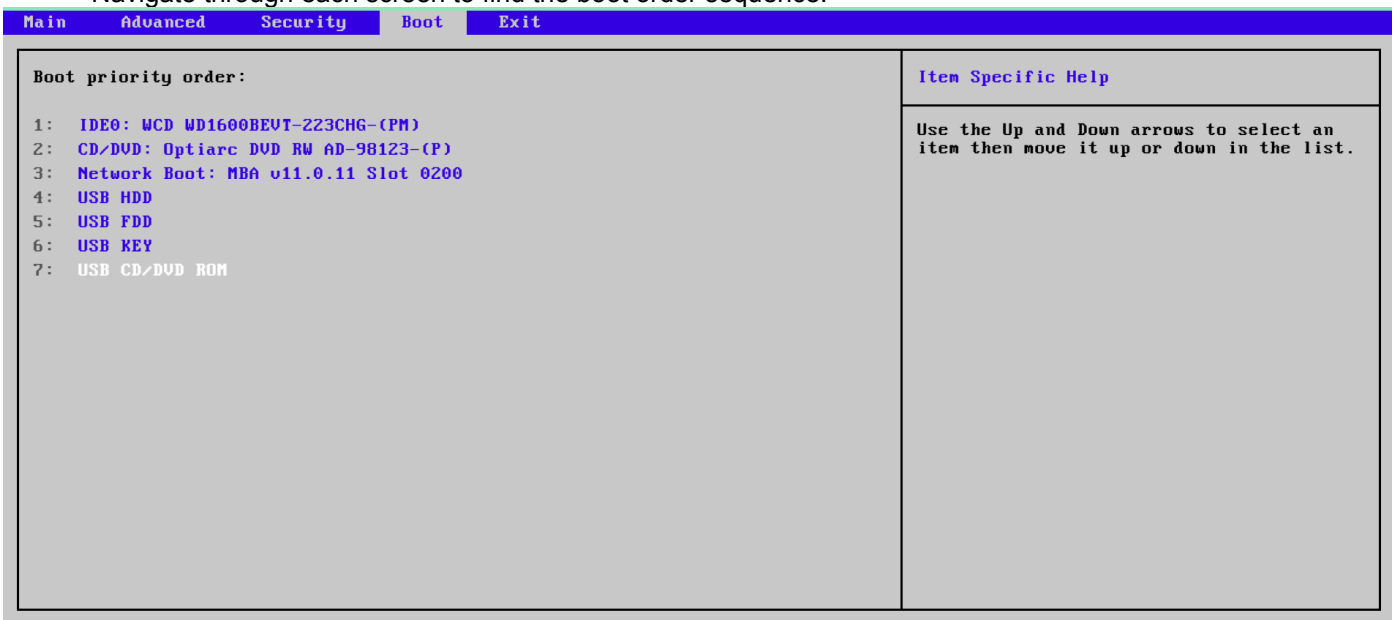
Questions:

What information is listed for the hard drive?

It includes the type of connector which is SATA and the storage of the HDD which is 2TB.

Step 6: Find the boot order sequence

Navigate through each screen to find the boot order sequence.



Questions:

What is the first boot device in the boot order sequence?

The first to boot is the IDE Hardrive.

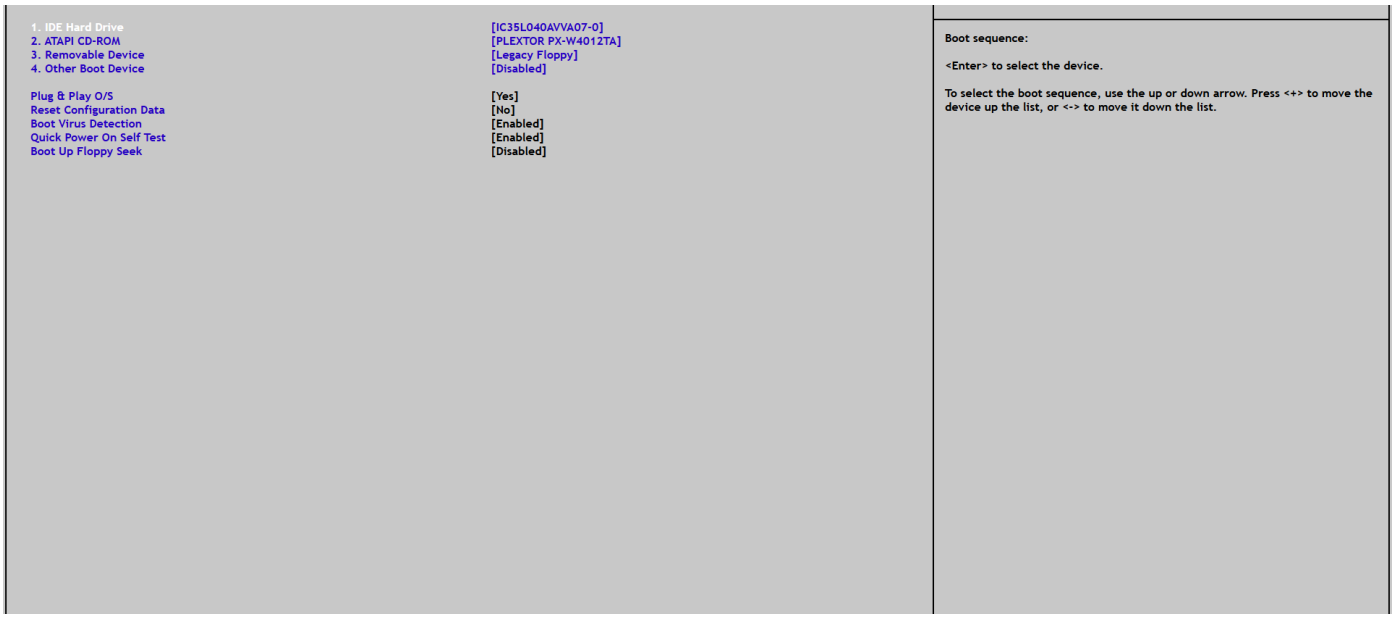
How many additional devices can be assigned in the boot order sequence?

A total of 7 can be assigned in the boot order sequence.

Step 7: Set the device boot order settings.

a. Ensure that the first boot order device is the optical drive.

b. Ensure that the second boot order device is the hard disk drive.



Questions:

Why would you change the first boot device to the optical drive?

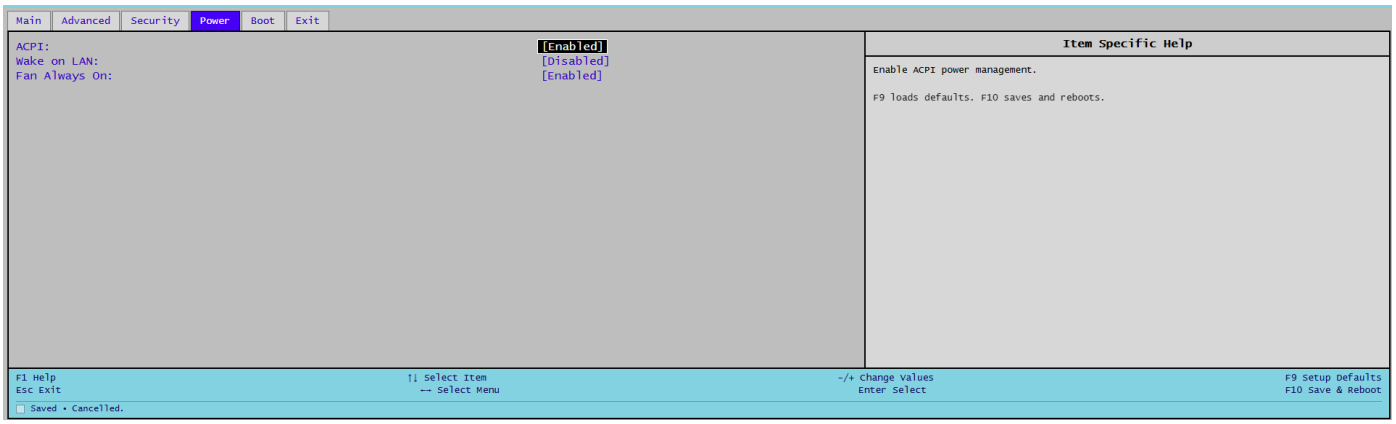
I would change the first boot device to the optical drive so the computer can start from a CD or DVD, especially when installing or repairing an operating system.

What happens when the computer boots and the optical drive does not contain bootable media?

If there is no bootable CD or DVD in the optical drive, the computer will check the next boot device, which is the hard disk drive, and start the operating system from there.

Step 8: Find the power management setup or ACPI screen.

Navigate through each screen to find the power management setup screen, or ACPI screen.

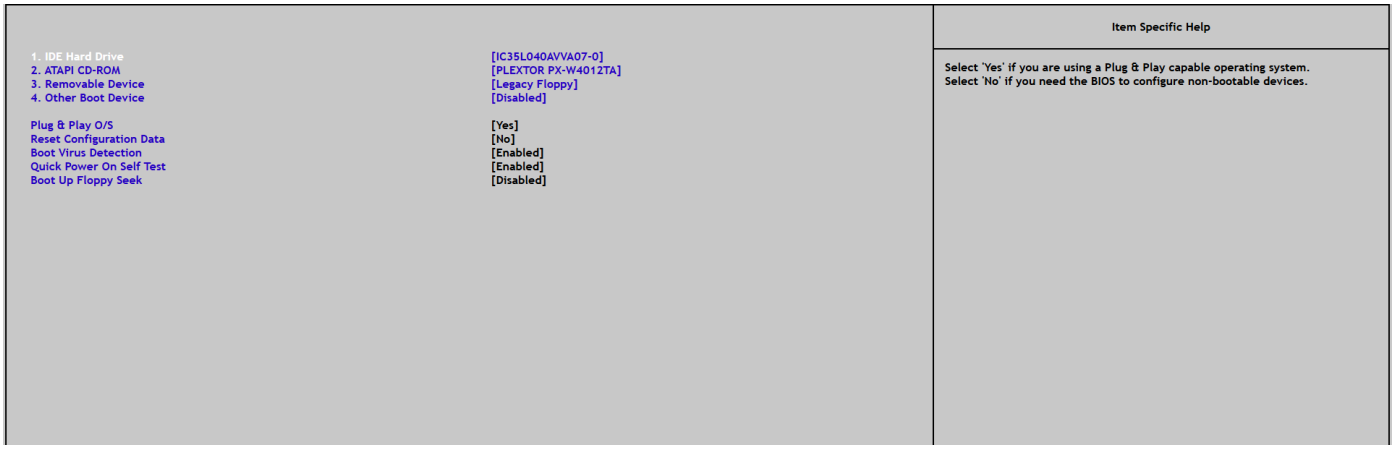


Question:

What power management settings are available?
Available power management options include the enable or disable ACPI, Wake on LAN, Fan always on.

Step 9: Find the PnP settings.

Navigate through each screen to find the PnP settings.



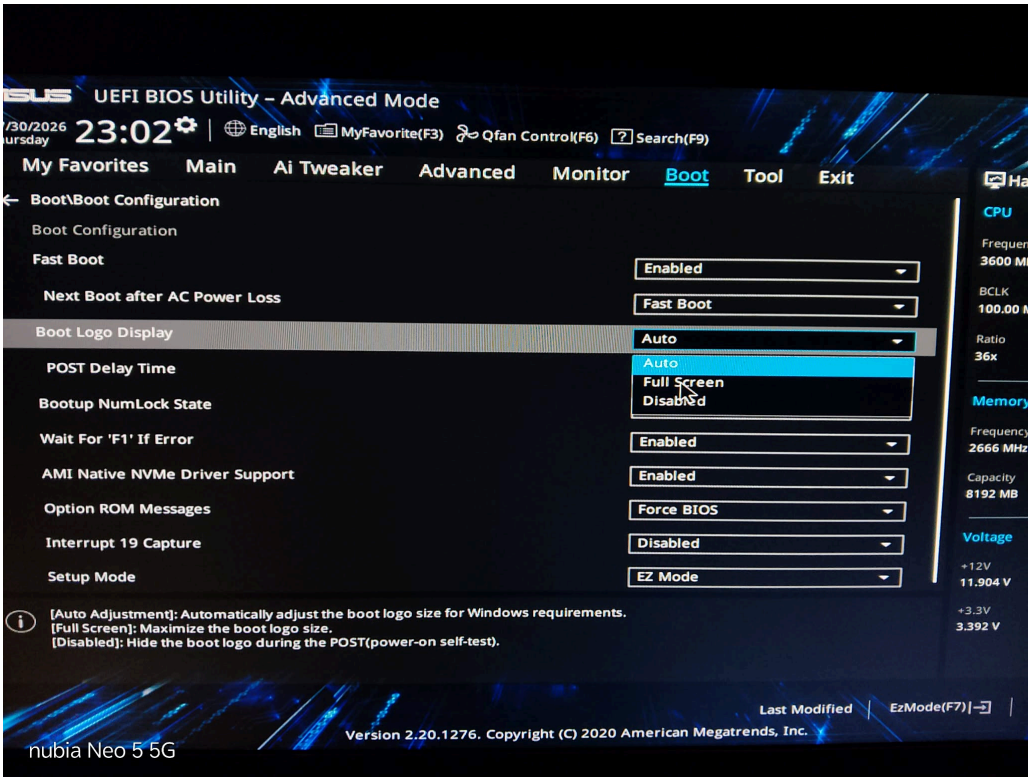
Question:

What PnP settings are available?

Available settings enable and disable the Plug n Play OS.

Step 10: Find the splash screen settings.

Navigate through each screen to find the splash screen settings.



Question:

What splash screen settings are available?

Available settings are Auto, Fullscreen and Disabled.

Step 11: Save and exit the setup utility program.

Save the new BIOS/UEFI settings and exit the setup utility program. The computer should restart automatically.

Note: An error message stating that an OS cannot be found (or a similar error) will appear on the screen after the computer boots. An operating system must now be installed to prevent this error. It is safe to turn off the computer at this time.

This lab is complete. Please have the instructor verify your work.